

ABDERRAHMEN LAMLOUMI

Software Engineering Student

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EN/FR (Native), GE (A2)

Final-year Software Engineering student at **ENSIMAG** seeking a **six-month internship starting February 2027** to build maintainable, scalable software, with industrial experience in systems and data engineering.

Experience

Schneider Electric

Sep 2024 – Sep 2026

Junior Software & Data Engineer (Apprenticeship)

Grenoble, France & Bangalore, India

- Developed internal software tools within model-based environments for enterprise energy management, enhancing operational reliability.
- Spearheaded semantic modeling and automated label-tagging for EcoStruxure building equipment using RDF, OWL, and SHACL.
- Executed an international engineering mission in Bangalore (Jun – Sep 2026), aligning semantic interoperability and structured data workflows across global teams (Canada, France, India, Australia).

Société Générale

2022

RPA Intern

Paris, France

- Explored backend development, automation workflows, and data-driven problem solving in a financial environment.
- Gained hands-on exposure to software engineering practices, process optimization, and technical collaboration.

Projects

OrionDB Go, gRPC, Lock-free systems, LSM storage, Roaring Bitmaps

2026

- Built a high-performance experimental Go telemetry database built from first principles and designed for large-scale data ingestion (+40,000 concurrent req/s) and observability workloads by implemented lock-free MPMC ring buffers, bitmap-based indexing, and an LSM-style storage pipeline to improve throughput and memory efficiency.

Deca Compiler Java, ANTLR4, Compiler Design, JaCoCo

2025

- Co-built in a team of 8 a full compiler in Java for the Deca language covering semantic analysis, register allocation, parsing, and intermediate representation generation, utilizing automated testing and validation practices such as fuzzing to ensure an end-to-end compilation pipeline's correctness.

Operating-System Kernel C, Assembly, Operating Systems, Systems Programming

2026

- Implemented core OS functionality including process management, memory handling, interrupts, and device I/O.

RISC-V FPGA Processor VHDL, FPGA, RISC-V, Hardware Design

2024

- Designed and tested a RISC-V processor in VHDL with a focus on simulation, synthesis, and hardware validation.

Education

ENSIMAG - Grenoble INP

Sep 2024 – Present

Engineering degree in Software Engineering, Expected Jun 2026

Grenoble, France

- Relevant Coursework:** Distributed Systems, Operating Systems, Compilers, Advanced Databases, Concurrent Programming, Graph Theory

POLYTECH - Grenoble INP

Sep 2022 – Jun 2024

PEIP Preparatory Program (Ranked in top 5% of cohort)

Grenoble, France

- Completed a two-year intensive preparatory program focused on mathematics, science, and engineering fundamentals.

Technical Skills

Programming Languages

Go, Java, C, Python, TypeScript, JavaScript
SQL, Bash, VHDL, MIPS Assembly

Development Tools

Git, GitHub, GitLab CI/CD, Jenkins
Maven, Gradle, JIRA, Jama

Frameworks & Technologies

React, Next.js, Node.js, Express, Spring Boot
REST APIs, SQLite, MySQL, MongoDB, Docker

Core Concepts

Data Structures & Algorithms, System Architecture
Distributed Systems, Concurrency, Caching, OS